The Astrophysical Journal Letters, 954:L50 (15pp), 2023 September 10 https://doi.org/10.3847/2041-8213/acf472
© 2023. The Author(s). Published by the American Astronomical Society.
OPEN ACCESS

Living with a Red Dwarf: The Rotation–Age Relationships of M Dwarfs

Scott G. Engle and Edward F. Guinan
Villanova University, Dept. of Astrophysics and Planetary Science, 800 E. Lancaster Ave., Villanova, PA 19085, USA
Received 2023 June 7; revised 2023 July 24; accepted 2023 August 28; published 2023 September 12

Abstract

Age is a fundamental stellar property, yet for many stars, it is difficult to reliably determine. For M dwarfs, it has been notoriously so. Due to their lower masses, core hydrogen fusion proceeds at a much slower rate in M dwarfs than it does in more massive stars like the Sun. As a consequence, more customary age determination methods (e.g., isochrones and asteroseismology) are unreliable for M dwarfs. As these methods are unavailable, many have searched for reliable alternatives. The M dwarfs comprise the overwhelming majority of the nearby stellar inventory, which makes the determination of their fundamental parameters even more important. Further, an ever-increasing number of exoplanets are being found to orbit M dwarfs, and recent studies have suggested that they may have a relatively higher number of low-mass planets than other spectral types. Determining the ages of M dwarfs then allows us to better study any hosted exoplanets as well. Fortunately, M dwarfs possess magnetic activity and stellar winds like other cool dwarf stars. This causes them to undergo the spin-down effect (rotating with longer periods) as they age. For this reason, stellar rotation rate has been considered a potentially powerful age determination parameter for over 50 yr. Calibrating reliable age–rotation relationships for M dwarfs has been a lengthy process, but here we present the age–rotation relationships for ∼M0–6.5 dwarfs, determined as part of the Living with a Red Dwarf program. These relationships should prove invaluable for a wide range of stellar astrophysics and exoplanetary science applications.

Unified Astronomy Thesaurus concepts: Stellar ages (1581); Stellar rotation (1629); Low mass stars (2050); Photometry (1234); Late-type dwarf stars (906); M dwarf stars (982); White dwarf stars (1799); Stellar evolution (1599)

1. Introduction and Background: Studying M Dwarfs

Main-sequence (dwarf) M stars (dM stars, red dwarfs; hereafter referred to as M dwarfs) represent the cool, low-mass, low-luminosity end of the main sequence and comprise ∼75% of all stars in the solar neighborhood (Reylé et al. 2021). This study specifically focuses on M0V–∼M6.5V stars with properties of mass ≈0.6–0.1 $M_\odot$, radius ≈0.6–0.1 $R_\odot$, luminosity ≈0.06–0.001 $L_\odot$, and temperature $T_{\rm eff} = 3900$–2850 K.[1]

The M dwarfs have received substantial attention during the 21st century, prompted in part by the discovery that these numerous stars host a relatively large number of terrestrial-sized planets (France et al. 2020; Hsu et al. 2020; Rojas-Ayala 2023) when compared to stars of higher mass. Aside from the large number of nearby M dwarfs available for study, they also make very attractive targets for terrestrial planet searches and research programs, as such planets are more readily detected through radial velocity motions and planetary transits due to the low masses and small radii of the M dwarf host stars. Estimates of the frequency of potentially habitable planets hosted by M dwarfs have been made primarily from Kepler Mission data but also from numerous radial velocity studies. Conservative estimates place the planetary frequency around 15% (Dressing & Charbonneau 2013), and studies including expanded circumstellar habitable zone (HZ) estimates indicate higher frequencies of ∼30%–40% (Hsu et al. 2020; Kopparapu 2013). If a slightly conservative "middle ground" of 25% is adopted, it implies that within 10 pc (∼33 lt-yr) of the Sun (a volume of space containing ∼240 M dwarfs), there should be ∼60 potentially habitable Earth-sized planets. Extrapolating to include the entire Milky Way raises the possibility that billions of Earth-sized planets are orbiting within the HZs of M dwarfs.

The M dwarfs are equally fascinating targets for stellar astrophysics. Relative to their sizes, they display enhanced magnetic dynamo activity due to the extent of their interior convective zones. As a result, their coronal and chromospheric emissions are also relatively strong compared to their bolometric luminosities. They have comparatively slow core nuclear reaction rates, however, which makes them rather "fuel-efficient" and results in their long main-sequence lifetimes. More massive M dwarfs can live on the main sequence for over 100 Gyr, while those of lower mass ($M < 0.2\ M_\odot$) can live as long as ∼1 trillion (∼10^{12}) yr (Choi et al. 2016). Due to this, no M dwarfs have yet to evolve off the main sequence. Another consequence of their long lifetimes, however, is that once M dwarfs reach the core hydrogen-fusing main sequence, their basic physical properties (L, $T_{\rm eff}$, and R) remain essentially constant over cosmological timescales (i.e., ∼14 Gyr).

Their large numbers, longevities, and near-constant main-sequence luminosities make M dwarfs very compelling targets for programs searching for life in the Universe, since, unlike our Sun, the HZs and thus exoplanet bolometric irradiances (and planetary instellations) remain stable for tens of gigayears or longer. However, the stars' very slow nuclear evolution makes determining accurate stellar ages extremely challenging (see Soderblom 2010, and references therein).

Fortunately, it has been known for 50 yr (Skumanich 1972) that cool dwarfs undergo a "spin-down effect," whereby their

[1] https://www.pas.rochester.edu/~emamajek/EEM_dwarf_UBVIJHK_colors_Teff.txt

Original content from this work may be used under the terms of the Creative Commons Attribution 4.0 licence. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

rotation periods lengthen as they age. Since that time, numerous studies have shown the potential that stellar rotation holds as an age determinant—the method known as "gyrochronology" (Barnes 2003, 2007; Mamajek & Hillenbrand 2008; Engle & Guinan 2011, 2018; Pass et al. 2022). Late F, G, K, and M dwarfs have stellar winds and magnetic fields that act in tandem to propagate the spin-down effect. The winds of these stars are magnetically threaded, continually carrying small amounts of each star's mass out into space while it is still (over a certain distance) tethered to the star itself by the magnetic field. The mass eventually escapes the magnetic field entirely, but its magnetically threaded tenure has already caused a slowing of the star's rotation due to conservation of angular momentum (Kawaler 1988).

This spin-down effect allows magnetic activity levels (observed through such proxies as X-ray and UV emissions and several emission features known to exist within optical spectra) to serve as additional age determinants for cool dwarfs. Activity–age relationships have also been constructed for M dwarfs, and they will be detailed in a follow-up paper.

The largest difficulty resides in building a representative sample of M dwarfs with a wide range of previously known ages and then determining their rotation periods. In this paper, we present these "benchmark" objects and the rotation–age relationships of M dwarfs determined as part of the Living with a Red Dwarf (LivRed) Program.

2. Dating M Dwarfs: Determining Ages for (Mostly) Ageless Stars

Age, along with mass and composition, is one of three key factors governing a star's current state (Soderblom 2010), yet it is also one of the most difficult stellar parameters to accurately measure. As mentioned, this is particularly true with M dwarfs, for which other commonly applied methods (e.g., isochronal, asteroseismic) for aging a star are unreliable (Lu et al. 2021). Observables, such as rotation period and X-ray and UV activity level, are known to be age-dependent and are often related to each other, but relating either quantity to stellar age first requires a set of M dwarfs with known ages—a benchmark sample.

With no currently available methods for directly determining the ages of single, isolated M dwarfs, the sample of benchmark M dwarfs has instead been built using age by association (see Tables 2 and 3). Each benchmark either has a stellar companion or belongs to a larger group or population of stars within the galaxy. For each pairing or grouping of stars, it is the age of the companion star or the group that can also be applied to the M dwarf, since they are assumed to have formed at the same time.

The age-by-association method in this study can be divided into three categories. For young dwarfs with ages below ∼2 Gyr, there are several well-studied "stellar groups" (referred to as either moving groups, clusters, or associations) available. The ages are very reliable but sadly do not cover nearly the range that we need. There are a limited number of additional clusters with greater ages, but other issues exist. For example, in the clusters NGC 752 and Ruprecht 147 (ages of 1.4, 2.5, and 2.7 Gyr; see Gruner & Barnes 2020; Curtis et al. 2020, and references therein), rotation periods have only been measured for earlier M dwarfs. A small number of HR 1614 moving group members (with ages of ∼2.0 Gyr—[60]) with rotation periods were once used, but the coherence of the moving group itself has recently been called into question (Kushniruk et al. 2020), which prompted their removal as benchmarks. Traditionally, the distances of highly prized targets such as M67 and NGC 188 (age of 4 Gyr) prevented sufficient time-series photometry of any of their faint M dwarf members. However, Dungee et al. (2022) recently measured rotation rates within this cluster for stars as late as ∼M3, showing some of the exciting recent progress in cluster gyrochronology measurements. See Section 2.2 for further details regarding M67.

Ages can also be assigned to stars that are members of specific galactic populations. Additional benchmarks were selected that belong to either the thick disk or halo populations (ages of ∼8–11 and ∼10–12.5 Gyr) of the Milky Way based primarily on the star's *UVW* galactic space motions (Leggett et al. 1998; Bensby et al. 2014), with further support of membership from metallicity values and velocity dispersions (Yu & Liu 2018). The advanced ages of these populations make them important benchmarks, but more direct age estimates for individual M dwarfs, as opposed to statistically supported, kinematically inferred ages, would usually be preferred.

The final and very welcome source of benchmarks is M dwarfs that belong to common proper motion (CPM) pairs/systems with an age-determinable companion. If the companion is a more massive (F–G dwarf) star, then a reliable age can be determined by other, more common (e.g., isochronal and/or asteroseismic) methods and applied to the (assumed to be coeval) M dwarf. Systems with white dwarf (WD) companions have become increasingly useful due to advances in determining the WD progenitor star properties (see Cummings et al. 2018) that have resulted in increasingly reliable ages. It is always important to note that the separation of the M dwarf from its companion is assumed to have prevented past interactions, allowing the M dwarf to evolve as if it were a single, isolated star, though for specific pairs (particularly those with small separations), the possibility of past interactions may exist (see Pass et al. 2022). However, a particular benefit these systems have over the previously mentioned CPM pairs is that the WDs do not outshine their M dwarf companions, which facilitates CCD photometry of the M dwarfs to search for rotation periods. These systems provided several M dwarf targets with ages older than 2 Gyr, an age range that was long awaiting additional targets. Determining rotation periods for these older M dwarfs became a primary focus of the program.

For this study, multiple (if available) measurements of each WD companion's effective temperature ($T_{\rm eff}$) and surface gravity (log g) were gathered from the recent literature, and mean values and uncertainties were determined via χ^2 analysis (Table 4). With these values, updated ages and uncertainties were calculated using both the `WD_Models` package, written by Dr. Sihao Cheng, and the `wdwarfdate` package, written by Dr. Rocio Kiman (Kiman et al. 2022). Both incorporate the latest WD cooling models available (Bédard et al. 2020) and the initial–final mass relationship (IFMR) of Cummings et al. (2018). We note that this is not the only IFMR choice available within the `wdwarfdate` package, but it is the one we selected both for consistency with `WD_Models` and because recent literature shows that it is still the most often used IFMR.

2.1. Calculating WD Ages

Using ages derived from WD companions is still something of a novel technique. The use of mean values for the WDs was deemed an important method for this program to mitigate the

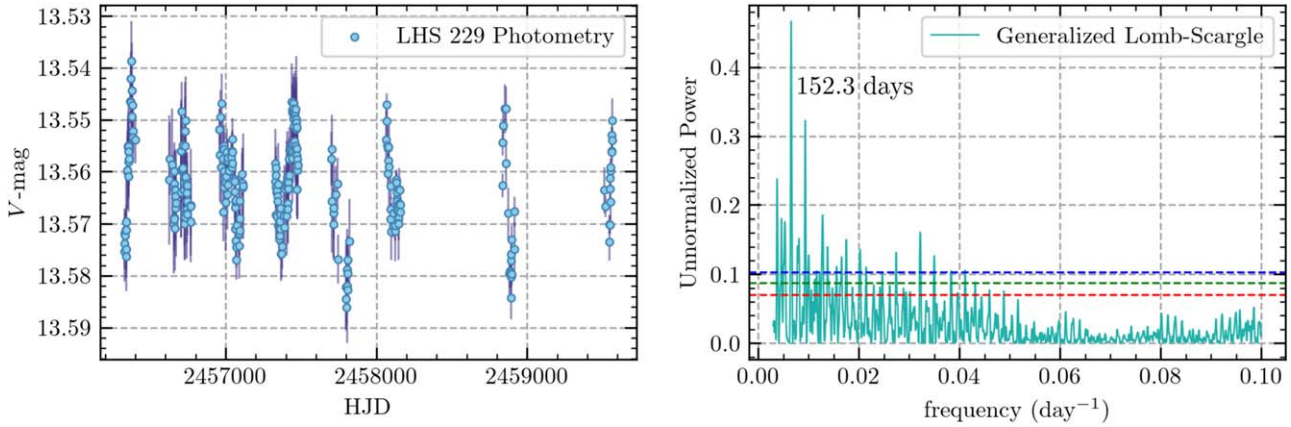

Figure 1. Example light curve for one of our mid–late benchmark stars, LHS 229, whose age of ∼8.7 Gyr (see Table 3) was determined from its WD companion, LHS 230 (see Table 4). At left, the full time-series data set is plotted, covering a span of 10 yr. At right, the generalized Lomb–Scargle periodogram (via `astropy`) results are plotted in frequency space, and the dashed horizontal lines indicate false-alarm probabilities of 10% (red), 1% (green), and 0.1% (blue). A rotation period of ∼152.3 days was found, and the phased light curve is plotted in Figure 3.

Table 1
Rotations Periods in Addition to Those Found in the Literature for NGC 752 and Ruprecht 147 Members

Star Name	P_{rot} (days)
NGC 752	
2MASS J01561789+3757110	11.0
2MASS J01572038+3747404	18.2
2MASS J01583743+3750269	12.6
2MASS J01584750+3734036	10.6
2MASS J01570853+3750213	50.9
2MASS J01572291+3757466	8.7
2MASS J01570025+3747243	25.0
2MASS J01582469+3733380	21.7
Ruprecht 147	
Gaia DR2 4179511879236506112	27.1
Gaia DR2 4199836897135629440	28.4
Gaia DR2 4180583692619791360	20.0
Gaia DR2 4184987580281615616	24.9
Gaia DR2 4184292792011931776	32.6

Note. All rotation periods have uncertainties of 0.1 day or below, but we are reporting the values to this level of precision until the periods are confirmed through dedicated follow-up photometry.

uncertainties that could arise from choosing any single measure for individual targets, especially since no single study could be found that had measured every WD we wished to analyze. To give some estimate of this effect, for all benchmarks included in Gentile Fusillo et al. (2021), ages computed using their mean properties were compared to those computed using only the Gentile Fusillo et al. (2021) values. An average age difference of ∼0.8 Gyr was found, though a major contributor is LP 498-26, for which Gentile Fusillo et al. (2021) determined a fairly low log g value. If only LP 498-26 is removed from the comparison, the average age difference drops to ∼0.5 Gyr.

Additionally, as mentioned at the end of Section 2, the `wdwarfdate` package offers more than one IFMR. We chose to use that of Cummings et al. (2018) for the reasons mentioned, although an alternative exists (Marigo et al. 2020) that uses newer data and appears to show a new feature in the relationship. Again, age comparisons were made, but this time between those determined using the two different IFMRs. An average age determination of ∼0.2 Gyr was found.

2.2. The Complicated Case of M67

Though plotted in Figures 5 and 6, the data for M67 were not included when determining our relationships. Dungee et al. (2022) quoted an age of 4 Gyr, which is the often-quoted age within the literature. However, the full age range quoted in Dungee et al. (2022) is 3.5–5 Gyr, and isochronal studies perhaps point more toward an age of ∼3.5 Gyr (e.g., Gruner et al. 2023). A number of gyrochronology studies appear to arrive at or near 4 Gyr, but we felt that perhaps it would not be appropriate to use a primarily gyrochronology-based age when calibrating the rotation–age relationships presented here.

Using the relationships and rotation values listed here, we determine an early M dwarf age for M67 of $\sim 3.2 \pm 0.8$ Gyr and a mid–late M dwarf age of $\sim 3.6 \pm 1.0$ Gyr.

3. Staring at M Dwarfs: Determining Rotation Periods

The surface features (e.g., starspots) of cool, main-sequence stars will be brought in and out of view as the stars rotate if the orientation of the star (the inclination of the star's rotation axis relative to our line of sight) and the starspot surface distribution are favorable. Repeatedly measuring stellar brightness via photometry can determine the rotation periods by revealing cyclical changes in brightness over time. This was the preferred method for determining benchmark rotation periods, as it is precise and works for very long rotation periods where spectroscopic measurements of rotation velocity become ambiguous. Measuring rotation via photometry is a very straightforward process on paper, but in practice, substantial difficulties can arise when dealing with M dwarfs. As the LivRed program was designed to study the crucial missing age range of >3 Gyr, it was unknown at the outset, but this would mean measuring rotation periods anywhere from ∼30 days to as long as ∼150–170 days. Such extended rotation periods require rather lengthy observing campaigns. As light amplitudes can be below ∼0.015 mag, the photometry requires a sufficiently high precision as well. Further, successfully detecting a rotation signal depends on the star maintaining a "favorable" (nonuniform) distribution of starspots, possibly for several years. Fortunately, many of the M dwarfs observed in

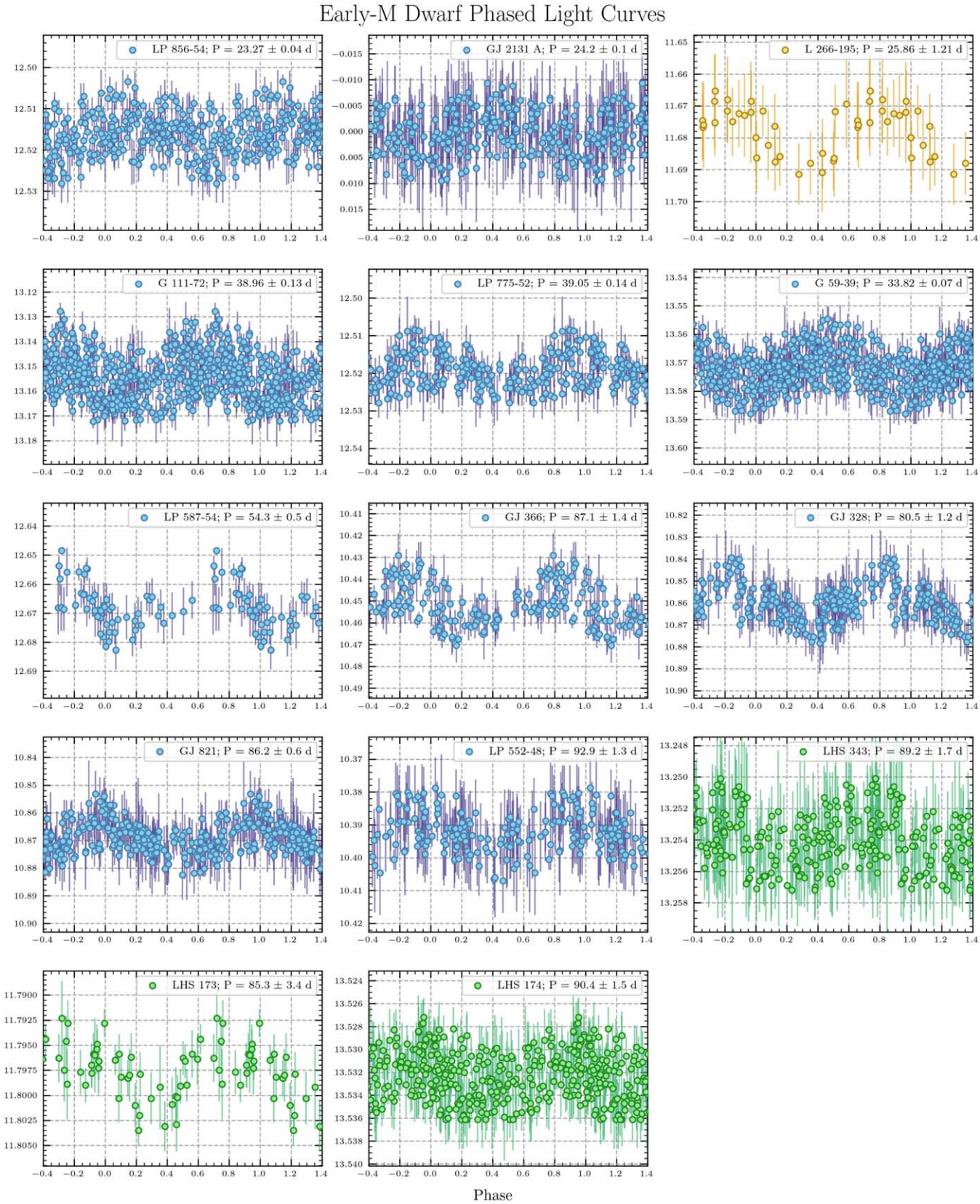

Figure 2. Phased rotation light curves of the early M dwarf benchmarks. The name and rotation period (and uncertainty) of each star is inset within their respective plot. The "scatter" of the data along the *y*-axis is primarily due to variations in light-curve shape and amplitude over the different observing seasons. For plots where the magnitudes are centered on zero, a long-term trend or cycle was first removed from the data prior to conducting the rotation period search. Data are color-coded according to their source: blue for data that we acquired with the RCT (and APT in limited cases), green for public ZTF data (IRSA 2022), and beige for data we obtained with the Skynet telescope network.

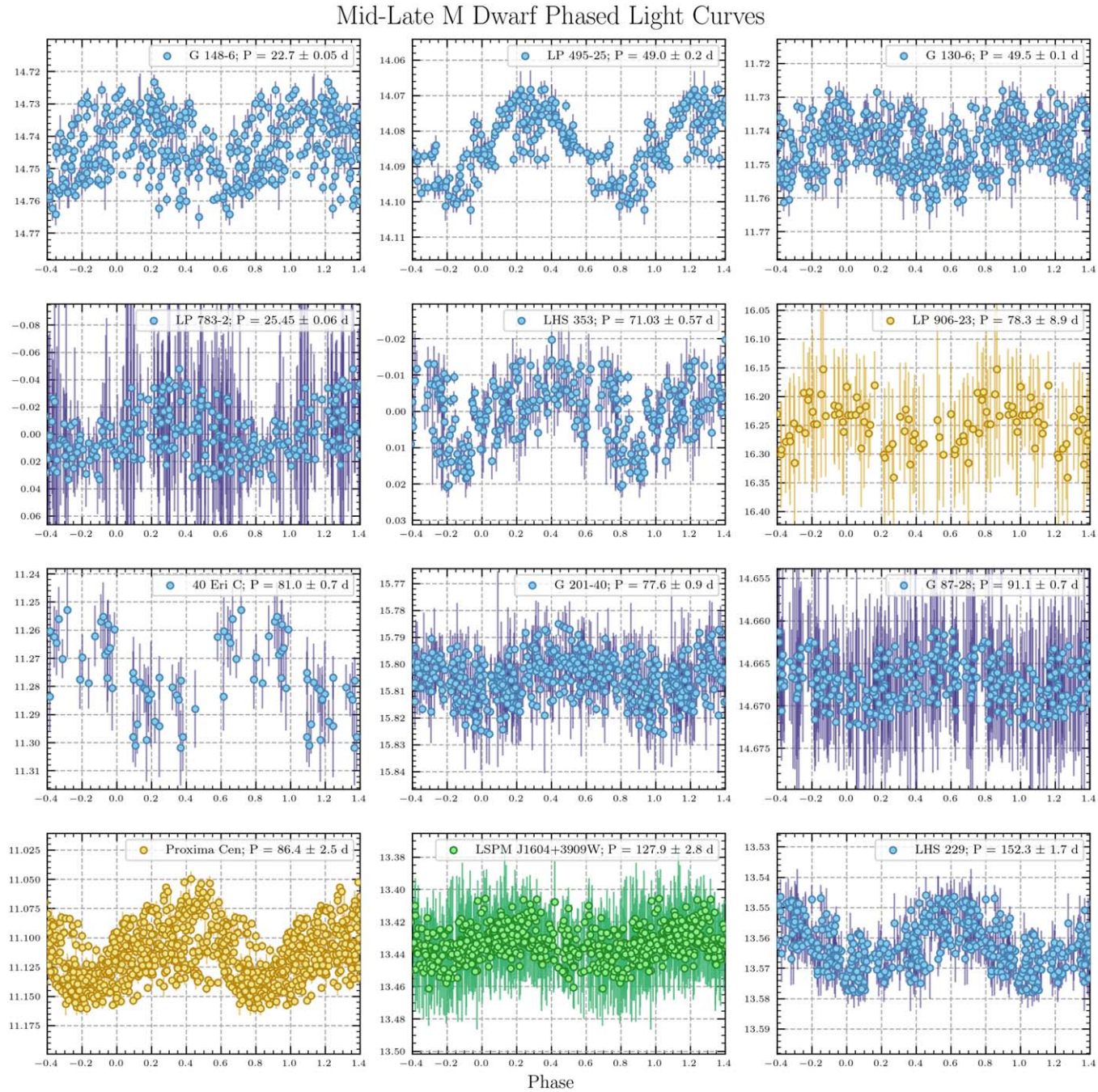

Figure 3. Phased rotation light curves of the mid–late M dwarf benchmarks (additional light curves are plotted in Figure 4). The data color scheme from Figure 2 is applied here, and, again as in Figure 2, a plot where the magnitudes are centered on zero indicates that a long-term trend or cycle has been removed.

this program have displayed a persistence of surface features, in some cases for several years. Though most of our targets are significantly older, our results thus far align with those of Robertson et al. (2020), who studied four young, rapidly rotating M dwarfs.

Apart from cluster members, whose rotation periods were obtained from the literature (see Tables 2 and 3), the vast majority of benchmark rotation periods were determined through dedicated CCD photometry of the targets carried out with the 1.3 m Robotically Controlled Telescope (RCT; Strolger et al. 2014) at Kitt Peak National Observatory in Arizona. In limited cases, data (or additional data) were obtained using other telescopes (e.g., the 0.8 m Automated Photoelectric Telescope, APT; see Engle 2015) or publicly available surveys either due to target visibility or to help confirm the rotation period. Except for the faintest sources, photometry was carried out in the V band, with individual measurement uncertainties of typically ∼0.004–0.01 mag depending on the brightness of the star. Data would be removed prior to analysis for reasons ranging from legitimate hardware malfunctions to poor sky quality conditions, even down to a large moth having taken a poorly timed stroll through the telescope's light path. For RCT/APT targets, three to five measurements were obtained per night, from which nightly means and uncertainties were determined. The yearly and multiyear data sets were searched for periodic variations

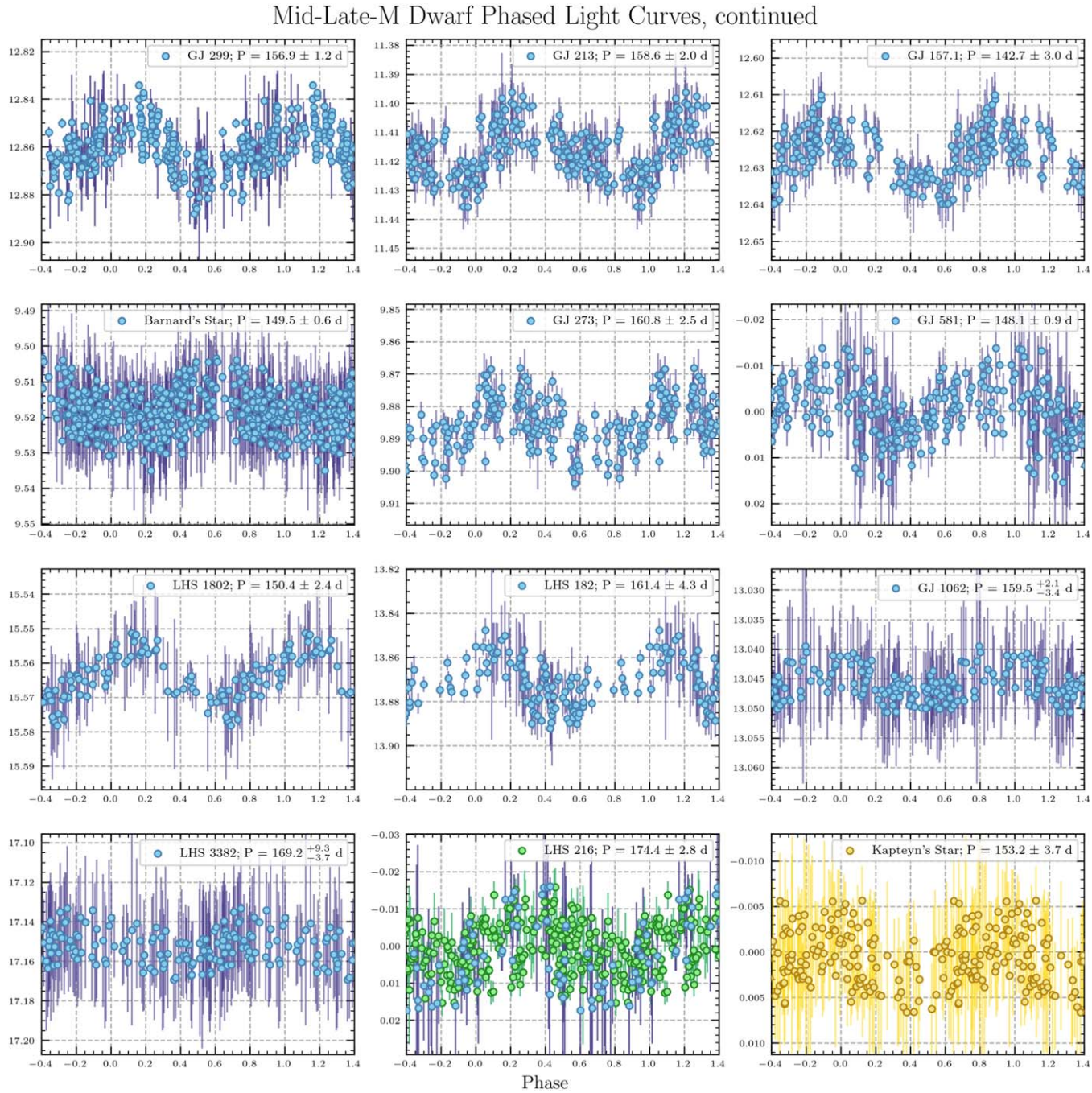

Figure 4. Additional phased rotation light curves of the mid–late M dwarf benchmarks, continued from and plotted in a similar fashion as Figure 2.

with the generalized Lomb–Scargle and CLEANest algorithms (as implemented within `astropy`, Astropy Collaboration et al. 2013, 2018, 2022; and `Peranso` (v3), Paunzen & Vanmunster 2016), an example of which is shown in Figure 1. All rotation signals have false-alarm probabilities below 1%.

In limited cases, additional or alternative sources of photometry were used to determine rotation periods. Proxima Cen and Kapteyn's Star have been a part of the program for several years, but they are Southern Hemisphere targets and thus inaccessible to the RCT. Observing time on the Skynet Robotic Telescope Network was purchased, and CCD photometry was carried out using the network's PROMPT telescopes at the Cerro Tololo Inter-American Observatory in Chile and the Meckering Observatory in Australia. Images were automatically reduced before downloading, and nightly means were determined and analyzed in similar fashion to RCT/APT data.

The remaining cases consist of targets that were added to the program only recently. Due to the installation and testing of a new and upgraded camera system and the Contreras wildfire, the RCT experienced an observing hiatus. With a good combination of photometric depth, precision, and timeline, Zwicky Transient Facility (ZTF) data were used as an alternative. Individual measurements were downloaded from the NASA/IPAC Data Archive, excluding any measurements flagged by the archive or obtained at high airmass. The resulting data were then sigma clipped (also within `astropy`) before nightly means were constructed. One example is the

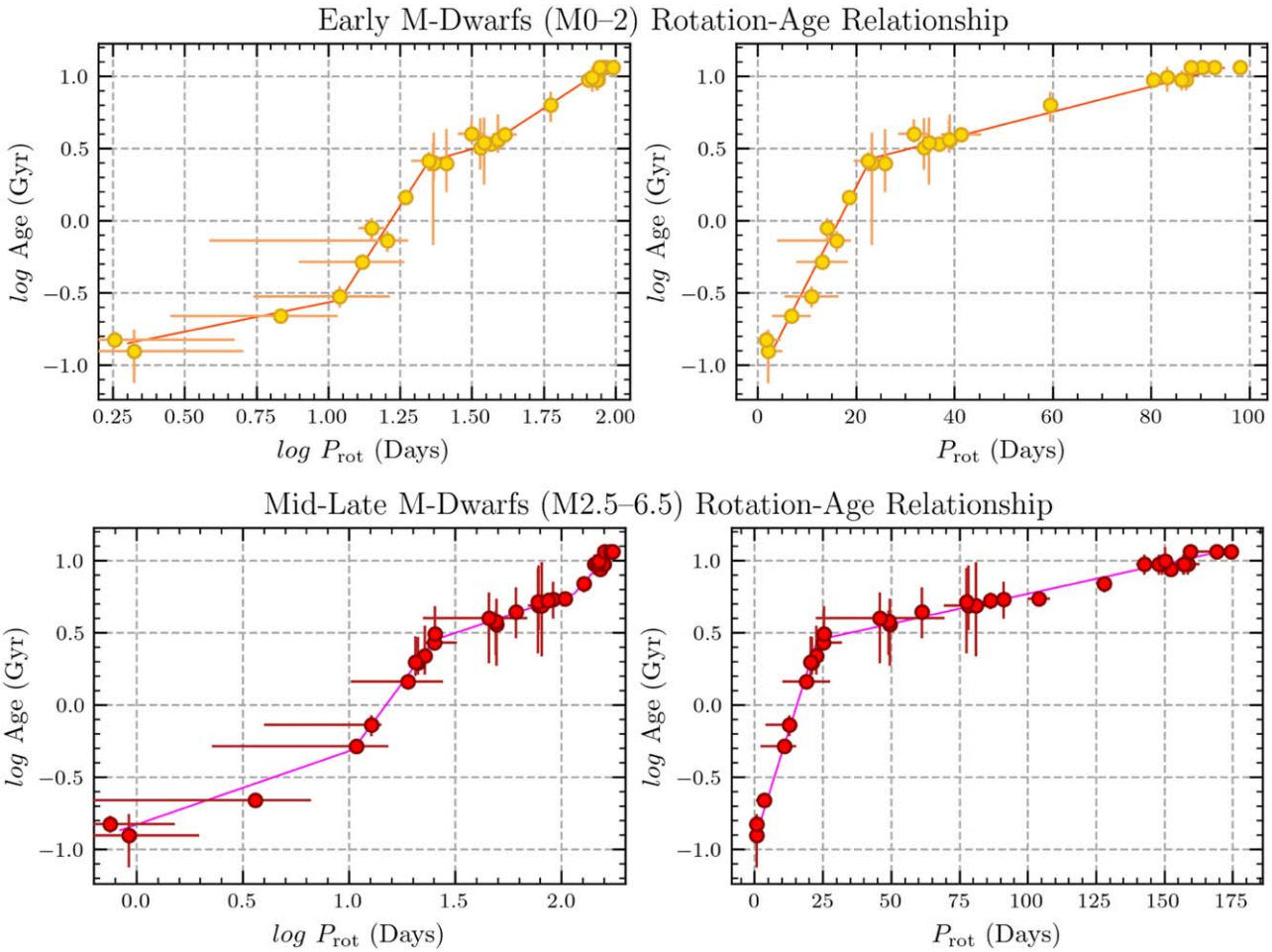

Figure 5. Plots showing log-log (left) vs. semi-log (right) age vs. rotation relationships for the early (top) and mid–late (bottom) M dwarfs. As shown in the figure, the data of both groups are better linearized in semi-log form. It is perhaps possible that each "segment" of the log-log plots represents a real evolutionary stage—e.g., the first segment is pre-main-sequence evolution, and the second represents the interiors of the M dwarfs synchronizing—and the semi-log plots merge these two distinct stages. However, given the current data and spread of rotation rates in the clusters, such a firm conclusion cannot be drawn here.

open cluster NGC 752, where ZTF photometry was used to both confirm the rotation periods of Agüeros et al. (2018) and provide additional periods (see Table 1). Similarly, we were able to measure rotation periods for a small sample of Ruprecht 147 mid–late M dwarfs using ZTF photometry. Members of both clusters are now being observed with the RCT to further confirm the previous results and, more specifically, to expand the inventory of reliably determined rotation periods.

The rotational light curves of the benchmark M dwarfs are shown in Figures 2–4 to give an idea of the amplitudes of variability observed and the data quality. The M dwarf rotation amplitudes can range from ∼0.01 mag or less in the "toughest" cases to as much as 0.06 mag for ideal cases such as Proxima Cen.

Several additional potential benchmarks have been observed, and continue to be, but are not included in this study. Some simply have yet to show a coherent rotation period. For others (e.g., LHS 3279), more than one potential rotation period is observed, and further data will hopefully reveal the true rotation period. There are others still for which rotation periods have been found that are at odds with the currently determined age, but perhaps with reason. One example is G121-21, which has an age (determined via its WD companion) of ∼2 Gyr, yet RCT photometry revealed a rotation period of ∼0.62 days, which we confirmed by analyzing TESS photometry. Though it is possible that this star represents an extreme example of the spread of rotation periods at younger ages, we believe that an unresolved companion orbiting the M dwarf is more likely to be the cause of its rapid rotation.

4. Results and Discussion: A Tale of Two Relationships

The primary focus of the Living with a Red Dwarf (LivRed) program has been to characterize the evolution of M dwarf rotation rates over their lifetimes, with the end goal of providing a reliable method for calculating the age of an M dwarf as long as its rotation period has been determined. When comparing related, age-associated quantities (e.g., activity and rotation), the data are commonly linearized by taking the logarithms of both quantities. In our analysis of M dwarf age versus rotation, we found that both subsets of M dwarfs (but particularly the mid–late subset) showed deviations from linearity in log-log space, and a more straightforward analysis of their rotations over time could be carried out in semi-log space (see Figure 5) while clearly showing the inflection points on the evolutionary tracks of both early and mid–late M dwarfs that will be discussed later in this section.

As mentioned previously, constructing these relationships proved a rather complicated task, but not simply due to the difficulty in building a substantial set of benchmark targets and

Table 2
The Early M Dwarf (M0–2) Benchmarks

Name	Sp. Type	Age (Gyr)	P_{rot} (days)	Age Determined via
Pleiades	M0–2V	0.125 [+0.05, −0.05]	2.11 [+2.96, −1.50][a]	Cluster
NGC 2516	M0–2V	0.15 [+0.02, −0.02]	1.80 [+2.93, −1.30][a]	Cluster
M34	M0–2V	0.22 [+0.02, −0.02]	6.82 [+3.99, −3.99][a]	Cluster
NGC 3532	M0–2V	0.30 [+0.05, −0.05]	10.96 [+5.47, −5.47][b]	Cluster
M37	M0–2V	0.52 [+0.06, −0.06]	13.15 [+5.24, −5.24][a]	Cluster
Praesepe/Hyades	M0–2V	0.73 [+0.12, −0.12]	16.06 [+2.91, −12.2][c]	Cluster
NGC 6811	M0–2V	0.89 [+0.15, −0.15]	14.19 [+1.45, −1.45][d]	Cluster
NGC 752	M0–2V	1.46 [+0.18, −0.18]	18.6 [+4.2, −4.2][e]	Cluster
L266-195	M1V	2.50 [+1.82, −0.93]	23.19 [+1.84, −1.84]	WD comp (L266-196)
LP 856-54	M1–1.5V	2.52 [+1.58, −0.61]	23.27 [+0.04, −0.04]	WD comp (LP 856-53)
Ruprecht 147	M0–2V	2.6 [+0.4, −0.4]	22.4 [+2.9, −2.9][f,g]	Cluster
G59-39	M0V	3.20 [+1.96, −0.93]	33.82 [+0.07, −0.07]	WD comp (EGGR 92)
LP 775-52	M0–1V	3.39 [+0.29, −0.20]	36.90 [+0.10, −0.10]	WD comp (LP 775-53)
GJ 2131 A	M1V	3.47 [+1.69, −1.69]	34.82 [+0.10, −0.10]	WD comp (GJ 2131 B)
G111-72	M1.5–2V	3.64 [+1.35, −0.68]	38.96 [+0.13, −0.13]	WD comp (G111-71)
HIP 43232 B	M1.5V	3.95 [+0.35, −0.35][h]	41.3 [+4.1, −4.1][h]	MS comp (HIP 43232 A)
M67	M0–2V	4.0 [+1.0, −0.5]	31.8 [+3.3, −3.3][i]	Cluster
LP 587-54	M1.5–2V	6.34 [+1.49, −1.38]	59.5 [+1.4, −1.4]	WD comp (LP 587-53)
GJ 366	M1.5V	9.5 [+1.5, −1.5]	87.1 [+1.2, −1.2]	Thick disk/halo population
GJ 328	M0V	9.5 [+1.5, −1.5]	80.5 [+0.6, −0.6]	Thick disk/halo population
GJ 821	M1V	9.5 [+1.5, −1.5]	86.2 [+1.1, −1.1]	Thick disk/halo population
LP 552-48	M0V	9.88 [+1.78, −2.06]	83.2 [+0.8, −0.8]	WD comp (LP 552-49)
LHS 343	sdK7	11.5 [+1.0, −1.5]	89.2 [+1.7, −1.7]	Halo population
LHS 173	esdK7[j]	11.5 [+1.0, −1.5]	85.3 [+3.4, −3.4]	Halo population
LHS 174	sdM0[j]	11.5 [+1.0, −1.5]	90.4 [+1.5, −1.5]	Halo population

Notes.
[a] Godoy-Rivera et al. (2021).
[b] Fritzewski et al. (2021).
[c] Núñez et al. (2022).
[d] Curtis et al. (2019).
[e] Agüeros et al. (2018), with additional rotation periods measured by us using ZTF photometry.
[f] Curtis et al. (2020).
[g] 2020AA644A16G.
[h] Sawczynec, E. (2021), thesis available at https://www.phys.hawaii.edu/wp-content/uploads/2021/06/ESawczynec_Thesis-1.pdf.
[i] Dungee et al. (2022).
[j] Kesseli et al. (2019).

the observational burden of measuring their rotation periods. When it comes to spin-down, it appears there is no way to broadly classify all M dwarfs; they represent too wide a range of parameters.

If you group G dwarfs together, the masses can vary by ∼10%. Grouping all K dwarfs together, the mass can vary by ∼30%–40%. But when studying M dwarfs, even focusing only on M0–6.5 dwarfs, as we have done here, the mass can vary by >500%. Important changes occur within the stars' interior structures and evolutionary timelines, which can be observed (and need to be accounted for) in the rotation relationships.

One dramatic difference was suspected early in the study but required lengthy follow-up. Preliminary project results showed that the rotations of (especially older) early versus mid–late M dwarfs followed divergent evolutionary paths (Engle & Guinan 2018) as they aged. This splitting of M dwarf subsets into distinct rotation-based groups has been observed in other studies as well (see Popinchalk et al. 2021). Also particularly relevant to this study: models have shown (Louis Amard, private communication) that the oldest stars (subdwarf members of the halo population) display an interesting and related phenomenon in which their interiors are structured as that of a main-sequence star with a slightly later spectral type (see Tables 2 and 3). An example is Kapteyn's Star, which would initially be considered a member of our early subset, since it is classified as sdM1.5, yet models indicate it has a fully convective interior similar to an ∼M2.5 or later main-sequence star (Louis Amard, private communication). A potential explanation for subdwarfs having deeper convective zones than their spectral types would indicate is that their smaller radii lead to larger interior temperature gradients, but confirming the true cause requires further study.

By an age of 10 Gyr, the average mid–late M dwarf will have a rotation period almost twice as long as the average early M dwarf (∼155 versus ∼85 days; see Figures 5 and 6). These different paths resulted in our first subdivision of the M dwarfs into what we call the "early" (M0–2) and "mid–late" (M2.5—6.5) groups. This is near but earlier than the usual spectral type of M3–3.5, which is routinely quoted as the transition point to a fully convective interior. Mullan & Houdebine (2020) recently showed, however, that changes due to magnetic effects (which would be rotation-related) likely occur within the M2.1–2.3 range, which is encouraging in light of our rotation period results. The decision to not include stars with spectral types later than ∼M6.5V is intentional. First, we presently do not have a sufficient sample of older stars at these later spectral types with both well-determined ages and rotation periods. Second, from the small sample of such stars that is available, it

Table 3
The "Mid" M Dwarf (M2.5–6.5) Benchmarks

Name	Sp. Type	Age (Gyr)	P_{rot} (days)	Age Determined via
Pleiades	M2.5–∼3.5V[a]	0.125 [+0.05, −0.05]	0.92 [+1.06, −0.43][b]	Cluster
Pleiades	M4–∼6.5V[a]	0.125 [+0.05, −0.05]	0.38 [+0.15, −0.10][b]	Cluster
NGC 2516	M2.5–∼3.5V	0.15 [+0.02, −0.02]	0.68 [+0.77, −0.30][b]	Cluster
NGC 2516	M4–6V	0.15 [+0.02, −0.02]	0.41 [+0.09, −0.10][b]	Cluster
M34	M2.5–∼3.5V	0.22 [+0.02, −0.02]	3.63 [+3.01, −3.01][b]	Cluster
M37	M2.5–6V	0.52 [+0.06, −0.06]	10.84 [+4.55, −8.57][b]	Cluster
Praesepe/Hyades	M2.5–∼3.5V[a]	0.73 [+0.12, −0.12]	12.81 [+1.42, −8.83][c]	Cluster
Praesepe/Hyades	M4–∼6.5V[a]	0.73 [+0.12, −0.12]	23.8 [+3.3, −3.3][c]	Cluster
NGC 752	M2.5–∼3.5V	1.46 [+0.18, −0.18]	19.83 [+10.93, −10.93][d]	Cluster
NGC 752	M4–6V	1.46 [+0.18, −0.18]	13.7 [+5.8, −9.8][d]	Cluster
NSV 11919	M2.5–3V	1.97 [+0.98, −0.35]	21.07 [+0.05, −0.05]	WD comp (NSV 11920)
Gaia DR3 7552[5a]	M3–3.5V	1.98 [+1.01, −0.39]	20.6 [+1.1, −1.1]	WD comp (Gaia DR3 4784[5b])
G148-6	M3–3.5V	2.20 [+1.35, −0.57]	22.7 [+0.05, −0.05]	WD comp (G148-7)
Ruprecht 147	M2.5–∼4V	2.6 [+0.4, −0.4]	26.6 [+4.2, −4.2][f]	Cluster
LP 783-2	M6.5V	3.12 [+1.73, −0.56]	25.45 [+0.06, −0.06]	WD comp (LP 783-3)
G130-6	M3V	3.60 [+1.84, −1.73]	49.5 [+0.1, −0.1]	WD comp (G130-5)
LP 498-25	M2.5V	3.80 [+0.70, −1.58]	49.0 [+0.2, −0.2]	WD comp (LP 498-26)
M67	M2–∼3.5V	4.0 [+1.0, −0.5]	46.0 [+23.5, −23.5][g]	Cluster
LHS 353	M4V	4.42 [+2.13, −1.52]	61.2 [+2.1, −2.1]	WD comp (GJ 515)
LP 906-23	M4V	4.86 [+4.41, −1.57]	78.3 [+8.9, −8.9]	WD comp (LP 906-24)
40 Eri C	M4.5V	4.89 [+4.81, −2.71]	81.0 [+0.7, −0.7]	WD comp (40 Eri B)
G201-40	M3–3.5V	5.16 [+3.77, −2.87]	77.6 [+0.9, −0.9]	WD comp (G201-39)
Proxima Cen	M5.5V	5.3 [+0.7, −0.7]	86.4 [+2.5, −2.5]	α Cen system
G87-28	M4V	5.42 [+1.70, −1.45]	91.1 [+0.7, −0.7]	WD comp (G87-29)
2MASS J23095781+5506472	M5.5–6V	5.47 [+0.61, − 0.26]	104.1 [–, –]	WD comp (2MASS J23095848+5506491)
LSPM J1604+3909W	M4V	6.9 [+0.9, −0.9]	127.9 [+2.8, −2.8]	MS comp (HD 144579)
LHS 229	M4V	8.71 [+0.56, −0.46]	152.3[+1.7, −1.7]	WD comp (LHS 230)
GJ 299	M4.5V	9.5 [+1.5, −1.5]	156.9 [+1.2, −1.2]	Thick disk/halo population
GJ 213	M4V	9.5 [+1.5, −1.5]	158.6 [+2.0, −2.0]	Thick disk/halo population
GJ 157.1	M4V	9.5[+1.5, −1.5]	142.7 [+3.0, −3.0]	Thick disk/halo population
Barnard's Star (GJ 699)	M4V	9.5 [+1.5, −1.5]	149.1 [+0.6, −0.6]	Thick disk/halo population
GJ 273	M3.5V	9.5 [+1.5, −1.5]	157.3 [+5.7, −5.7]	Thick disk/halo population
GJ 581	M3V	9.5 [+1.5, −1.5]	148.1 [+0.9, −0.9]	Thick disk/halo population
LHS 1802	M5V	9.97 [+1.98, −2.39]	150.4 [+2.4, −2.4]	WD comp (LHS 1801)
LHS 182	usdM0	11.5 [+1.0, −1.5]	161.4 [+5.2, −5.2]	Halo population
GJ 1062	sdM2.5	11.5 [+1.0, −1.5]	159.5 [+2.1, −3.4]	Halo population
LHS 3382	esdM2.5	11.5 [+1.0, −1.5]	169.2 [+9.3, −3.7]	Halo population
LHS 216	esdM3[h]	11.5 [+1.0, −1.5]	174.4 [+2.8, −2.8]	Halo population
Kapteyn's Star (GJ 191)	sdM1.5[h]	11.5 [+1.0, −1.5]	153.2 [+3.7, −3.7]	Halo population

Notes.
[a] Depending on the stellar parameter used, the rotation periods in these clusters were measured for objects as late as either M6 or M6.5.
[b] Godoy-Rivera et al. (2021).
[c] Núñez et al. (2022).
[d] Agüeros et al. (2018), with additional rotation periods measured by us using ZTF photometry.
[f] Curtis et al. (2020)
[5a] Full name: Gaia DR3 5172481276951287552.
[5b] Full name: Gaia DR3 5172481203936294784.
[g] Dungee et al. (2022).
[h] Classified using Gizis (1997) and Zhang et al. (2019).

appears they experience either no appreciable spin-down effect or one that is altogether different from any of the other M dwarf subsets presented here. A well-known example is the M8V star Trappist-1, which has an age of 7.6 ± 2.2 Gyr and a rotation period of 3.39 days (e.g., Burgasser & Mamajek 2017).

There is an additional complication at young ages due to the length of time required for each star to reach the main sequence. As with other issues, this one is also particularly difficult for M dwarfs, whose pre-main-sequence lifetimes can range from ∼140 Myr to ∼1.5 Gyr (M0 6.5 dwarfs; Choi et al. 2016). Due to this, on top of the spread of initial rotation rates normally displayed by all cool dwarf stars, younger M dwarfs show a larger range of rotation periods, as detailed in several excellent studies (see Douglas et al. 2019; Gruner & Barnes 2020; Curtis et al. 2020; Godoy-Rivera et al. 2021, for recent examples). After an age of ∼2.5–3 Gyr, all mid–late M dwarfs have converged onto a single evolutionary track, and any differences between their rotation-determined ages are negligible compared to the uncertainties of the relationships. At young ages, though, this group appears to require further subdivision, most likely as a result of lengthening pre-main-sequence lifetimes. Due to the distances of some of these young

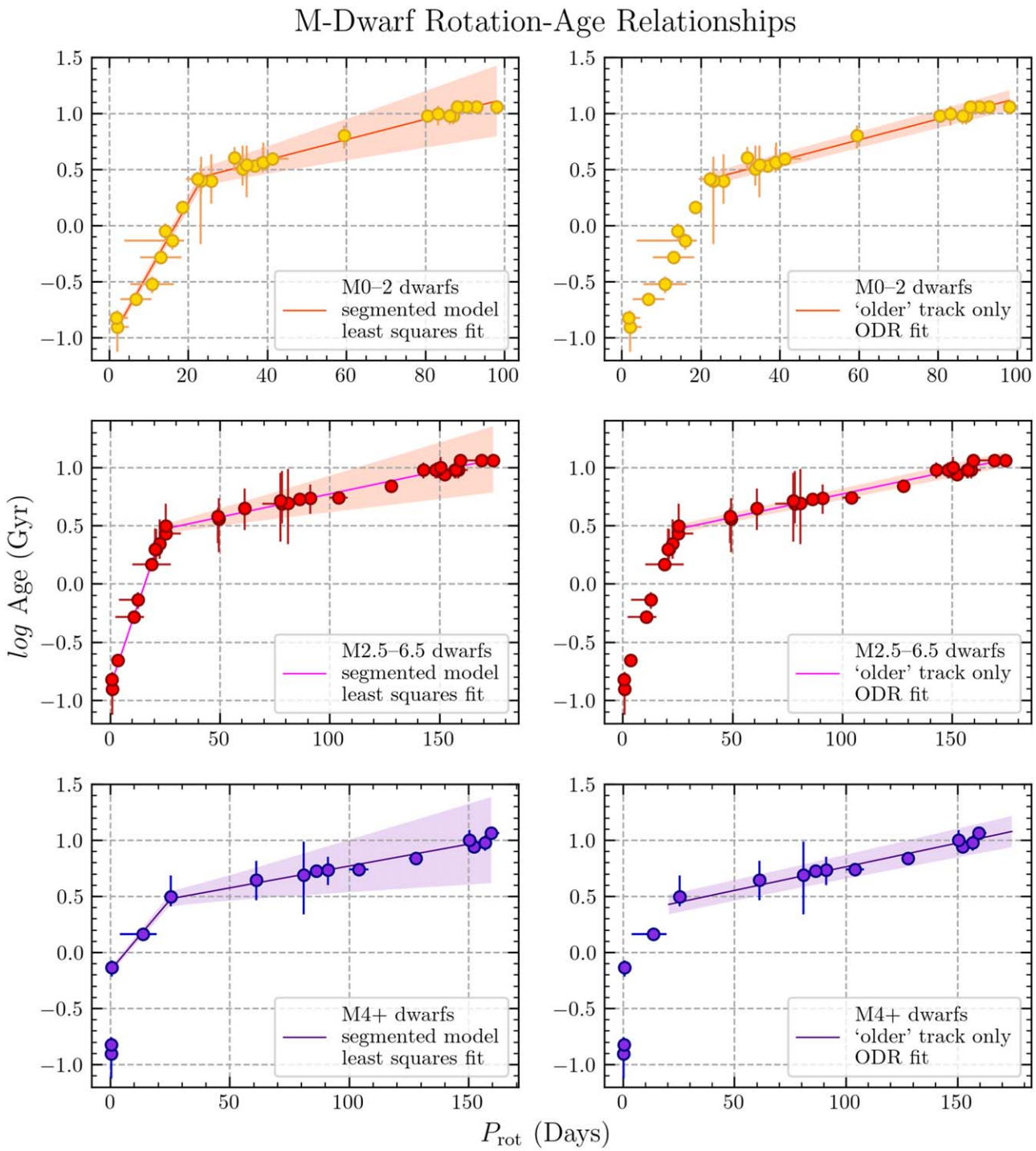

Figure 6. Age–rotation relationships are plotted for the different M dwarf subsets in semi-log space. The left panels show the segmented linear fit for each stellar subset's full time span of data, while the right panels show the single linear fit of only the subset's older evolutionary track. In the top panels, only the early (M0–2) subset is plotted. In the middle panels, the mid–late subset is plotted. This subset contains M2.5–6.5 dwarfs, except on the young track, where only M2.5–3.5 dwarfs are plotted. In the bottom panels, only the data for M4 and later spectral types are plotted. Clear inflection points can be seen in the plots of all stellar subsets. This is believed to occur when the interiors of the stars resynchronize, and the next phase of their rotational evolution can begin. Also note the drastically different rotation scales for each subset, with the early M dwarfs slowing to 100 days but the mid–late M dwarfs slowing much further, to 175 days.

clusters, this aspect of the study is in need of further study, but M dwarfs later than ∼M4 sensibly appear to follow a different evolutionary path while young (the middle and bottom panels of Figure 6 show this further subdivision).

Previous studies (see Curtis et al. 2020) have proposed that the rotation periods of cool dwarfs do not follow one continuous evolutionary track, characterized by a single power-law relationship ($P_{rot} \propto \text{age}^n$), with a "braking index" determined by the exponent (n). This was originally proposed by Skumanich (1972), with an initially determined value of $n = 0.5$ for that study's target sample, and recent studies have revised this value. Douglas et al. (2019) and Dungee et al. (2022), for example,

Table 4
Parameters Used to Derive Ages for the WD Companions

WD Name	WD Type	Model	log g	$T_{\rm eff}$	Source(s)
LP 856-53	DA5	H thick	8.06 [+0.05, −0.05]	9869 [+92, −92]	6, 17, 18, 19
GJ 2131 B	DA3.9	H thick	7.98 [+0.04, −0.04]	12,573 [+616, −616]	6, 9, 17
G111-71	DA6.5	H thick	8.04 [+0.02, −0.02]	7592 [+35, −35]	4, 6, 9, 18
LP 775-53	DA	H thick	8.16 [+0.05, −0.05]	6509 [+92, −92]	6, 18
EGGR 92	DA4	H thick	8.01 [+0.04, −0.04]	10,590 [+65, −65]	3, 6, 17, 18, 21
LP 587-53	DA8.6	H thick	7.97 [+0.02, −0.02]	5792 [+48, −48]	3, 4, 6
LP 552-49	DC	H thin	7.72 [+0.4, −0.4]	4431 [+145, −145]	4, 13
NSV 11920	DBZ5	H thin	8.105 [+0.02, −0.02]	11,070 [+96, −96]	2, 7, 11, 16, 28, 29
Gaia DR3 4784	non-DA	H thin	8.15 [+0.05, −0.05]	9960 [+112, −112]	28, 29
G148-7	DA3.1	H thick	8.02 [+0.02, −0.02]	15,964 [+104, −104]	3, 6, 9, 13, 18, 26, 27
LP 783-3	DZ6.5	H thin	8.10 [+0.03, −0.03]	7924 [+97, −97]	6, 11, 16, 21
G130-5	DA4	H thick	8.01 [+0.01, −0.01]	12,682 [+58, −58]	3, 6, 9, 13, 18
LP 498-26	DB3	H thin	8.01 [+0.04, −0.04]	15,868 [+257, −257]	5, 6, 7
GJ 515	DA4	H thick	7.96 [+0.03, −0.03]	14,346 [+258, −258]	9, 17, 19, 21, 28
40 Eri B	DA2.9	H thin	7.94 [+0.04, −0.04]	17,043 [+453, −453]	8, 9, 20
G201-39	DA5.6	H thick	7.95 [+0.03, −0.03]	9002 [+61, −61]	3, 6, 10, 17, 18
G87-29	DQ8	H thin	8.00 [+0.03, −0.03]	6802 [+79, −79]	4, 6, 13, 24, 25
LHS 230	DA+DA	H thick	8.11 [+0.01, −0.01]	4955 [+63, −63]	12, 13, 21
LHS 1801	DA	H thick	7.926 [+0.01, −0.01]	5146 [+22, −22]	4, 6, 13
L266-196	DA4.3	H thick	8.05 [+0.03, −0.03]	10,503 [+442, −442]	17, 28, 29
LP 906-23	DA7	H thick	7.985 [+0.09, −0.09]	6731 [+205, −205]	28, 29
2MASS J23095848+5506491	DA8.8	H thick	8.116 [+0.15, −0.15]	5552 [+28, −28]	1, 7, 13, 28

References. [1] McCleery et al. (2020); [2] Klein et al. (2020); [3] Kilic et al. (2020); [4] Blouin et al. (2019); [5] Kong et al. (2019); [6] Gentile Fusillo et al. (2019); [7] Rolland et al. (2018); [8] Bond et al. (2017); [9] Bédard et al. (2017); [10] Anguiano et al. (2017); [11] Subasavage et al. (2017); [12] Holberg et al. (2016); [13] Limoges et al. (2015); [14] Genest-Beaulieu & Bergeron (2014); [15] Kawka & Vennes (2012); [16] Giammichele et al. (2012); [17] Gianninas et al. (2011); [18] Garcés et al. (2011); [19] Koester et al. (2009); [20] Holberg et al. (2008b); [21] Holberg et al. (2008a); [22] Holberg & Bergeron (2006); [23] Castanheira et al. (2006); [24] Dufour et al. (2005); [25] Bergeron et al. (2001); [26] Bergeron et al. (1995); [27] Bergeron et al. (1992); [28] Gentile Fusillo et al. (2021);[29] Jiménez-Esteban et al. (2023).

each derived an index of $n \approx 0.62$, although Douglas et al. studied only F and G dwarfs—more massive than the stars studied here—but Dungee et al. included stars up to ∼M3.

For these data and the activity–age relationships in the following subsections, two fits were applied. A two-segment linear equation was defined using the `numpy.piecewise` function and fit to the full age range using `scipy.optimize.least_squares`, and an unsegmented linear model was fit to only the "older" track of each data set via orthogonal distance regression (ODR; using the `BCES`[2] module written by Rodrigo Nemmen). With fewer terms, and fitting only the time span after all mid–late M dwarfs appear to have converged onto a single evolutionary path, the unsegmented model shows significantly reduced uncertainties (as seen in the following equations and the ages calculated in Table 5). The final fitted age–rotation relationships are shown in Figure 6 with the best-fitting parameters as follows.

M0–2 dwarfs, segmented fit of full age range:

$$\begin{aligned}\text{logAge (Gyr)} &= 0.0621[0.0024] \times P_{\rm rot}\ \text{(days)} \\ &\quad -1.0437[0.0380] \\ &\text{for } P_{\rm rot} < 23.4933[0.7643] \\ \text{logAge (Gyr)} &= 0.0621[0.0024] \times P_{\rm rot}\ \text{(days)} \\ &\quad -1.0437[0.0380] \\ &\quad - 0.0528[0.0025] \times (P_{\rm rot} - 23.4933[0.7643]) \\ &\text{for } P_{\rm rot} \geqslant 23.4933[0.7643]. \end{aligned} \tag{1}$$

M0–2 dwarfs, linear ODR fit to the "older track" only:

$$\begin{aligned}\text{logAge (Gyr)} &= 0.0094[0.0002] \times P_{\rm rot}\ \text{(days)} \\ &\quad + 0.1909[0.0114] \\ &\text{for } P_{\rm rot} \gtrsim 22. \end{aligned} \tag{2}$$

M2.5–6.5 dwarfs (M2.5—M3.5 for $P_{\rm rot} \lesssim 24$):

$$\begin{aligned}\text{logAge (Gyr)} &= 0.0561[0.0012] \times P_{\rm rot}\ \text{(days)} \\ &\quad - 0.8900[0.0185] \\ &\text{for } P_{\rm rot} < 24.1888[0.4268] \\ \text{logAge (Gyr)} &= 0.0561[0.0012] \times P_{\rm rot}\ \text{(days)} \\ &\quad - 0.8900[0.0185] \\ &\quad - 0.0521[0.0012] \times (P_{\rm rot} - 24.1888[0.4268]) \\ &\text{for } P_{\rm rot} \geqslant 24.1888[0.4268]. \end{aligned} \tag{3}$$

M2.5–6.5 dwarfs, linear ODR fit to the "older track" only:

$$\begin{aligned}\text{logAge (Gyr)} &= 0.0041[0.0001] \times P_{\rm rot}\ \text{(days)} \\ &\quad +0.3691[0.0126] \\ &\text{for } P_{\rm rot} \gtrsim 22. \end{aligned} \tag{4}$$

M4–6.5 dwarfs:

$$\begin{aligned}\text{logAge (Gyr)} &= 0.0251[0.0018] \times P_{\rm rot}\ \text{(days)} \\ &\quad -0.1615[0.0303] \\ &\text{for } P_{\rm rot} < 25.4500[1.9079] \\ \text{logAge (Gyr)} &= 0.0251[0.0018] \times P_{\rm rot}\ \text{(days)} \\ &\quad -0.1615[0.0303] \\ &\quad - 0.0212[0.0018] \times (P_{\rm rot} - 25.4500[1.9079]) \\ &\text{for } P_{\rm rot} \geqslant 25.4500[1.9079]. \end{aligned} \tag{5}$$

[2] https://github.com/rsnemmen/BCES

Table 5
Rotation-based Age Determinations for Exoplanet-hosting M Dwarfs

Star Name	P_{rot}	P_{rot} Err.[a]	Age (Gyr) Segmented Fit	Err.[a]	Age (Gyr) ODR Fit	Err.[a]	Relationship Used	Note[b]
USco 1621 A	2.06	0.02	0.17	0.01	⋯	⋯	Mid–late	
HATS-74 A	4.75	0.05	0.18	0.02	⋯	⋯	Early	
AU Mic	4.86	0.01	0.18	0.02	⋯	⋯	Early	
COCONUTS-2 A	2.83	0.28	0.19	0.01	⋯	⋯	Mid–late	
USco 1556 A	4.67	0.05	0.24	0.01	⋯	⋯	Mid–late	
K2-284	8.88	0.4	0.32	0.04	⋯	⋯	Early	
TOI-620	8.99	0.09	0.33	0.03	⋯	⋯	Early	
K2-104	9.3	0.4	0.34	0.04	⋯	⋯	Early	
K2-240	10.8	0.1	0.42	0.05	⋯	⋯	Early	
Kepler-1512	10	0.01	0.47	0.02	⋯	⋯	Mid–late	
K2-141	14.03	0.09	0.67	0.08	⋯	⋯	Early	
Kepler-1410	14.09	0.7	0.68	0.11	⋯	⋯	Early	
TOI-540	0.73	0.01	0.72	0.05	⋯	⋯	M4+	
EPIC 211822797	14.6	1.1	0.73	0.14	⋯	⋯	Early	
TRAPPIST-1	1.4	0.05	0.75	0.05	⋯	⋯	M4+	
TOI-1227	1.65	0.04	0.76	0.05	⋯	⋯	M4+	
2MASS J04372171+2651014	1.84	0.02	0.77	0.05	⋯	⋯	M4+	
K2-25	1.88	0.01	0.77	0.05	⋯	⋯	M4+	
GJ 9066	1.96	0.02	0.77	0.05	⋯	⋯	M4+	
GJ 463	14	0.1	0.79	0.05	⋯	⋯	Mid–late	
HATS-76	15.16	0.2	0.79	0.10	⋯	⋯	Early	
Kepler-45	15.8	0.2	0.87	0.11	⋯	⋯	Early	
K2-415	4.3	0.06	0.88	0.06	⋯	⋯	M4+	
GJ 338 B	16.61	0.04	0.97	0.12	⋯	⋯	Early	*
Kepler-1229	17.63	0.88	1.13	0.21	⋯	⋯	Early	
GJ 685	18.15	0.15	1.21	0.16	⋯	⋯	Early	
Kepler-1455	18.32	0.92	1.24	0.23	⋯	⋯	Early	
K2-345	18.47	1.6	1.27	0.34	⋯	⋯	Early	
Gl 49	18.86	0.1	1.34	0.18	2.34	0.06	Early	*
TOI-1685	18.66	0.71	1.44	0.16	2.78	0.08	Mid–late	
Kepler-395	19.92	1	1.56	0.31	2.39	0.08	Early	
Kepler-705	20.09	1	1.60	0.32	2.40	0.08	Early	
TOI-1201	21	2	1.94	0.52	2.84	0.10	Mid–late	
GJ 3470	21.54	0.49	1.97	0.32	2.48	0.07	Early	
K2-264	22.8	0.6	2.36	0.42	2.55	0.08	Early	
TOI-3714	23.3	0.3	2.54	0.41	2.58	0.07	Early	
K2-286	23.8	3.7	2.63	0.53	2.61	0.22	Early	
BD-11 4672	25	0.3	2.69	0.51	2.67	0.08	Early	
Kepler-155	26.43	1.32	2.78	0.55	2.76	0.11	Early	
K2-95	23.9	2.4	2.83	0.91	2.92	0.11	Mid–late	
GJ 514	28	2.9	2.87	0.61	2.85	0.20	Early	
GJ 9827	28.72	0.19	2.92	0.60	2.90	0.08	Early	
GJ 96	29.6	2.8	2.97	0.65	2.96	0.20	Early	
L168-9	29.8	1.3	2.99	0.64	2.97	0.12	Early	
HD 147379	31	20	3.07	1.48	3.05	1.33	Early	
Kepler-1652	31.18	1.56	3.08	0.68	3.06	0.14	Early	
G9-40	29.85	1.01	3.09	0.34	3.08	0.10	Mid–late	
K2-332	31.71	3.6	3.15	0.37	3.14	0.14	Mid–late	
LP 714-47	33	3	3.20	0.76	3.18	0.23	Early	
GJ 393	34.15	0.22	3.28	0.78	3.26	0.10	Early	
TOI-776	34.4	1.4	3.30	0.79	3.28	0.14	Early	
K2-155	34.8	8.2	3.33	0.99	3.31	0.60	Early	
HATS-75	35.04	0	3.35	0.81	3.33	0.10	Early	
K2-18	38.6	0.5	3.35	0.45	3.34	0.10	Mid–late	
GJ 740	35.56	0.07	3.38	0.83	3.37	0.10	Early	
TOI-1759	35.65	0.17	3.39	0.83	3.37	0.10	Early	
GJ 720 A	36.05	1.38	3.42	0.85	3.40	0.14	Early	
GJ 3293	41	0.4	3.43	0.49	3.42	0.11	Mid–late	
HATS-71	41.72	0.14	3.45	0.50	3.44	0.11	Mid–late	
LSPM J2116+0234	42	2	3.46	0.51	3.45	0.12	Mid–late	
TOI-1468	42.5	1.5	3.47	0.51	3.46	0.12	Mid–late	
KOI-4777	44	1	3.52	0.54	3.51	0.11	Mid–late	
HD 260655	37.5	0.4	3.53	0.91	3.51	0.11	Early	

Table 5
(Continued)

Star Name	P_{rot}	P_{rot} Err.[a]	Age (Gyr) Segmented Fit	Err.[a]	Age (Gyr) ODR Fit	Err.[a]	Relationship Used	Note[b]
Gl 686	38.4	1.6	3.60	0.95	3.58	0.17	Early	
Kepler-235	39.49	1.97	3.68	1.00	3.67	0.19	Early	
GJ 9689	39.97	0.39	3.72	1.01	3.70	0.12	Early	
TYC 2187-512-1	40	1	3.72	1.02	3.71	0.14	Early	
K2-3	40	2	3.72	1.03	3.71	0.20	Early	
Kepler-560	50.47	2.52	3.74	0.66	3.73	0.15	Mid–late	
TOI-674	52	5	3.79	0.70	3.78	0.21	Mid–late	
TOI-700	54	0.8	3.86	0.72	3.85	0.13	Mid–late	
GJ 367	58	6.9	4.01	0.85	4.00	0.29	Mid–late	
TOI-1235	44.7	4.5	4.12	1.31	4.11	0.42	Early	
GJ 536	45.39	1.33	4.18	1.30	4.17	0.18	Early	
GJ 1252	64	4	4.23	0.96	4.22	0.21	Mid–late	
LHS 1678	64	22	4.23	1.28	4.22	0.87	Mid–late	
HD 180617	46.04	0.2	4.24	1.33	4.23	0.14	Early	
GJ 4276	64.3	1.2	4.25	0.95	4.24	0.15	Mid–late	
TOI-1695	47.7	2.2	4.40	1.44	4.38	0.25	Early	
LHS 1815	47.8	0.7	4.40	1.43	4.39	0.16	Early	
GJ 1265	70	0.7	4.47	1.10	4.46	0.15	Mid–late	
GJ 436	71.4	0.4	4.53	1.14	4.52	0.15	Mid–late	*
TOI-122	72	22	4.56	1.48	4.55	0.94	Mid–late	
L98-59	80.9	5	4.94	1.44	4.94	0.28	Mid–late	
TOI-2136	82.56	0.45	5.02	1.48	5.01	0.17	Mid–late	
YZ Cet	83	15	5.04	1.64	5.03	0.72	Mid–late	*
Teegarden's Star	86.3	1.3	5.20	1.60	5.19	0.19	Mid–late	*
Proxima Cen	86.4	2.5	5.20	1.61	5.19	0.22	Mid–late	*
GJ 3512	87	5	5.23	1.64	5.22	0.30	Mid–late	
GJ 411	56.15	0.27	5.27	2.03	5.27	0.18	Early	*
GJ 3323	88.5	0.89	5.30	1.68	5.30	0.19	Mid–late	*
GJ 3779	95	5	5.63	1.94	5.62	0.33	Mid–late	
G264-012	100	6	5.89	2.16	5.89	0.39	Mid–late	
TOI-237	102	22	6.00	2.53	6.00	1.24	Mid–late	
LTT 3780	104	15	6.11	2.46	6.11	0.88	Mid–late	
Wolf 1061 (GJ 628)	108.7	1.5	6.38	2.53	6.38	0.26	Mid–late	*
GJ 3929	122	13	7.21	3.34	7.22	0.91	Mid–late	
GJ 251	122.1	1.9	7.22	3.24	7.22	0.32	Mid–late	
GJ 1132	122.3	6	7.23	3.27	7.24	0.50	Mid–late	
Ross 128	123	1.2	7.28	3.29	7.28	0.30	Mid–late	
GJ 1214	124.7	5	7.40	3.41	7.40	0.45	Mid–late	
GJ 1002	126	15	7.48	3.62	7.49	1.08	Mid–late	
CD Cet	126.2	1.3	7.50	3.48	7.50	0.32	Mid–late	
GJ 486	130.1	1.6	7.77	3.73	7.78	0.34	Mid–late	
LHS 1140	131	1.3	7.84	3.79	7.84	0.34	Mid–late	
TOI-1634	77	26	8.27	6.44	8.29	4.70	Early	
GJ 625	77.8	5.5	8.41	4.70	8.43	1.07	Early	*
GJ 1151	140	10	8.51	4.49	8.52	0.87	Mid–late	
Wolf 1069	160	10	10.23	6.19	10.25	1.06	Mid–late	
GJ 273	160.8	2.5	10.31	6.20	10.33	0.54	Mid–late	*
GJ 3473	168.3	4.2	11.04	6.98	11.07	0.68	Mid–late	
HD 238090	96.7	3.7	12.64	8.81	12.72	1.19	Early	

Note.
[a] As explained in the text, uncertainties along the younger branch (age $\lesssim$ 2.9 Gyr) do not account for the full scatter of rotation rates in clusters and are therefore underestimated. Additionally, some rotation rates from the Exoplanet Archive did not have uncertainties.
[b] An asterisk in the "Note" column indicates that the rotation rate was updated to match the value presented either elsewhere in this paper or in the follow-up paper.

M4–6.5 dwarfs, linear ODR fit to the "older track" only:

$$\log\text{Age (Gyr)} = 0.0042[0.0002] \times P_{rot}\text{ (days)} + 0.3401[0.0288] \quad \text{for } P_{rot} \gtrsim 25. \tag{6}$$

Together, these relationships cover M0–6.5 dwarfs. Within 10 pc of the Sun, this represents ~72% of all stars with known spectral types (and ~48% of the wider range of objects, including stellar remnants and brown dwarfs; Reylé et al. 2021). As mentioned earlier in the paper, an increasing number of M dwarfs are being discovered as exoplanet hosts. Determining the ages of these stars, and thus the ages of their exoplanets, is important when selecting the ideal targets to study further for evidence of habitability or even life.

Single-celled organisms originated when the Earth (the only example we currently have for such events) was ∼0.7–0.9 Gyr old. The Great Oxygenation of the atmosphere occurred when Earth was ∼2.2 Gyr old, the Cambrian explosion and rapid diversification of complex life-forms occurred when Earth was ∼4–4.1 Gyr old, and technological civilization did not occur until the Earth was >4.5 Gyr old. Thus, exoplanet age is an important discriminator in the search for life. To demonstrate a benefit of our relationships, we provide the gyrochronological ages for all ∼M0–6.5 dwarf exoplanet hosts with a listed rotation period in the NASA Exoplanet Archive[3] in Table 5. The follow-up paper (Engle 2023) will focus on the X-ray and UV activity of M dwarfs over time and what insights these relationships offer for the stars, their magnetic dynamos, and their suitability to host potentially habitable planets.

We do wish to advise restraint, though, when using the first, or "young," tracks of the segmented relationships. These time spans were included to, as best we could, characterize the fullest evolutionary paths of average early to mid–late M dwarfs. However, as noted earlier in this section, M dwarfs display a wide range of rotation rates at young ages that are not represented by the relationship uncertainties (note the data point versus relationship uncertainties along the young tracks in Figure 6). Fortunately, estimates from activity–age relationships can extend the reliable range of age determinations into this younger time span, and this will be shown and discussed in the companion paper. Finally, while the segmented relationships determine young age uncertainties that are likely too small, including this time span ironically inflates the uncertainties of the older age range. For this time span, additional age and uncertainty measurements using fits to only the older data will be informative.

Barnes (2003) theorized that the rotational evolution of main-sequence stars showed two possible sequences, dependent on both time and mass (and therefore spectral type). The two sequences were termed the interface (I) and convective (C) sequences, named after the magnetic dynamos and interior structures of the stellar groups. The more massive, hotter stars spend only a short time on the C sequence before switching to the I sequence. On the I sequence, there is an interface between the radiative and convective regions of the stellar interior, but the magnetic field couples the regions together, and much of the stellar interior rotates as a rigid body. Lower-mass stars spend longer amounts of time on the C sequence before switching over, and fully convective stars likely never leave the C sequence. Spada & Lanzafame (2020) put forth an analytical model theorizing that stars do not initially evolve as rigid bodies. Rather, as the stellar surface loses angular momentum, a profile of differential rotation builds within the star. Angular momentum is transported from the interior to the surface, and eventually, the interior of the star "recouples." This accounts for the two-track evolutionary path, where the second evolutionary track begins after the interior of the star has recoupled. Spada & Lanzafame only calculated their model down to early M dwarf masses. At this mass range, however, the models predict that an ∼4 Gyr old early M dwarf will have an ∼31–32 day rotation period, where our data indicates an ∼40–45 day period.

To determine the braking indices of our M dwarf subsets and serve as an additional comparison to the literature results, rotation versus age data were fitted in linear space with a two-segment power-law equation. A best-fitting braking index (for the second evolutionary track) was determined to be 0.61 for the early M dwarfs and 0.62 for the mid–late M dwarfs—nearly identical to each other, well within the parameter uncertainties, and in excellent agreement with the results of Douglas et al. (2019) and Dungee et al. (2022). However, it is again worth noting that Douglas et al. (2019) based their braking index determination on solar-like F and G dwarfs and a comparison of the Praesepe cluster and the Sun. Just over 50 yr after Skumanich (1972) first discovered the spin-down effect operating in solar-type G dwarfs, it is an interesting implication that all cool dwarfs, from late F to ∼M6.5, may perhaps spin down according to the same braking index but simply with different recoupling timescales. Further investigation into the angular momentum loss of M dwarfs, using methods such as those of See et al. (2019) and Barnes & Kim (2010) and comparisons between measurements and estimated magnetic field strengths and mass-loss rates, are underway for inclusion in a follow-up paper.

These data and relationships will be of great use to the field and offer valuable insights into the most populous stellar members of our galaxy, M dwarfs. They allow for reliable ages and evolutionary histories to be determined but may also offer further insight into the differing dynamo mechanisms at work within the M dwarf subsets and how each mechanism influences or responds to the star's evolution over time.

Acknowledgments

We would like to acknowledge the tireless work of the RCT Consortium members, including former members David Laney and Louis-Gregory Strolger, for the operation and maintenance of the telescope and equipment, without which this study would not have been possible. We also wish to acknowledge the builders and operators of the Skynet telescope network and the Zwicky Transient Facility. Partial support for this project was provided by Chandra (CXO) grants GO0-21020X and AR8-19005X.

S.G.E. thanks Louis Amard for invaluable discussions and access to his M dwarf interior models. S.G.E. also thanks Andrej Prša for very helpful discussions regarding the regression fitting. We also achknowledge the valuable suggestions and comments made by an anonymous referee, which improved the overall quality of the paper.

Facilities: KPNO:RCT, CTIO:PROMPT.

Software: astropy (Astropy Collaboration et al. 2013, 2018, 2022), Matplotlib (Hunter 2007), Pandas (McKinney 2010), NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), AstroImageJ (Collins et al. 2017).

ORCID iDs

Scott G. Engle https://orcid.org/0000-0001-9296-3477
Edward F. Guinan https://orcid.org/0000-0002-4263-2650

References

Agüeros, M. A., Bowsher, E. C., Bochanski, J. J., et al. 2018, ApJ, 862, 33
Anguiano, B., Rebassa-Mansergas, A., García-Berro, E., et al. 2017, MNRAS, 469, 2102
Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167
Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., et al. 2018, AJ, 156, 123

[3] https://exoplanetarchive.ipac.caltech.edu/

Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, A&A, 558, A33
Barnes, S. A. 2003, ApJ, 586, 464
Barnes, S. A. 2007, ApJ, 669, 1167
Barnes, S. A., & Kim, Y.-C. 2010, ApJ, 721, 675
Bédard, A., Bergeron, P., Brassard, P., & Fontaine, G. 2020, ApJ, 901, 93
Bédard, A., Bergeron, P., & Fontaine, G. 2017, ApJ, 848, 11
Bensby, T., Feltzing, S., & Oey, M. S. 2014, A&A, 562, A71
Bergeron, P., Leggett, S. K., & Ruiz, M. T. 2001, ApJS, 133, 413
Bergeron, P., Liebert, J., & Fulbright, M. S. 1995, ApJ, 444, 810
Bergeron, P., Saffer, R. A., & Liebert, J. 1992, ApJ, 394, 228
Blouin, S., Dufour, P., Thibeault, C., & Allard, N. F. 2019, ApJ, 878, 63
Bond, H. E., Bergeron, P., & Bédard, A. 2017, ApJ, 848, 16
Burgasser, A. J., & Mamajek, E. E. 2017, ApJ, 845, 110
Castanheira, B. G., Kepler, S. O., Handler, G., & Koester, D. 2006, A&A, 450, 331
Choi, J., Dotter, A., Conroy, C., et al. 2016, ApJ, 823, 102
Collins, K. A., Kielkopf, J. F., Stassun, K. G., & Hessman, F. V. 2017, AJ, 153, 77
Cummings, J. D., Kalirai, J. S., Tremblay, P. E., Ramirez-Ruiz, E., & Choi, J. 2018, ApJ, 866, 21
Curtis, J. L., Agüeros, M. A., Douglas, S. T., & Meibom, S. 2019, ApJ, 879, 49
Curtis, J. L., Agüeros, M. A., Matt, S. P., et al. 2020, ApJ, 904, 140
Douglas, S. T., Curtis, J. L., Agüeros, M. A., et al. 2019, ApJ, 879, 100
Dressing, C. D., & Charbonneau, D. 2013, ApJ, 767, 95
Dufour, P., Bergeron, P., & Fontaine, G. 2005, ApJ, 627, 404
Dungee, R., van Saders, J., Gaidos, E., et al. 2022, ApJ, 938, 118
Engle, S. 2015, PhD thesis, James Cook Univ., Australia
Engle, S. G., & Guinan, E. F. 2011, in ASP Conf. Ser. 451, 9th Pacific Rim Conference on Stellar Astrophysics, ed. S. Qain et al. (San Francisco, CA: ASP), 285
Engle, S. G., & Guinan, E. F. 2018, RNAAS, 2, 34
France, K., Duvvuri, G., Egan, H., et al. 2020, AJ, 160, 237
Fritzewski, D. J., Barnes, S. A., James, D. J., & Strassmeier, K. G. 2021, A&A, 652, A60
Garcés, A., Catalán, S., & Ribas, I. 2011, A&A, 531, A7
Genest-Beaulieu, C., & Bergeron, P. 2014, ApJ, 796, 128
Gentile Fusillo, N. P., Tremblay, P. E., Cukanovaite, E., et al. 2021, MNRAS, 508, 3877
Gentile Fusillo, N. P., Tremblay, P.-E., Gänsicke, B. T., et al. 2019, MNRAS, 482, 4570
Giammichele, N., Bergeron, P., & Dufour, P. 2012, ApJS, 199, 29
Gianninas, A., Bergeron, P., & Ruiz, M. T. 2011, ApJ, 743, 138
Gizis, J. E. 1997, AJ, 113, 806
Godoy-Rivera, D., Pinsonneault, M. H., & Rebull, L. M. 2021, ApJS, 257, 46
Gruner, D., & Barnes, S. A. 2020, A&A, 644, A16
Gruner, D., Barnes, S. A., & Weingrill, J. 2023, A&A, 672, A159
Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Natur, 585, 357
Holberg, J. B., & Bergeron, P. 2006, AJ, 132, 1221
Holberg, J. B., Bergeron, P., & Gianninas, A. 2008a, AJ, 135, 1239
Holberg, J. B., Oswalt, T. D., Sion, E. M., & McCook, G. P. 2016, MNRAS, 462, 2295
Holberg, J. B., Sion, E. M., Oswalt, T., et al. 2008b, AJ, 135, 1225
Hsu, D. C., Ford, E. B., & Terrien, R. 2020, MNRAS, 498, 2249
Hunter, J. D. 2007, CSE, 9, 90
IRSA 2022, Time Series Tool, IPAC, doi:10.26131/IRSA538
Jiménez-Esteban, F. M., Torres, S., Rebassa-Mansergas, A., et al. 2023, MNRAS, 518, 5106
Kawaler, S. D. 1988, ApJ, 333, 236
Kawka, A., & Vennes, S. 2012, MNRAS, 425, 1394
Kesseli, A. Y., Kirkpatrick, J. D., Fajardo-Acosta, S. B., et al. 2019, AJ, 157, 63
Kilic, M., Bergeron, P., Kosakowski, A., et al. 2020, ApJ, 898, 84
Kiman, R., Xu, S., Faherty, J. K., et al. 2022, AJ, 164, 62
Klein, B., Blouin, S., Romani, D., et al. 2020, ApJ, 900, 2
Koester, D., Voss, B., Napiwotzki, R., et al. 2009, A&A, 505, 441
Kong, X., Luo, A. L., & Li, X.-R. 2019, RAA, 19, 088
Kopparapu, R. K. 2013, ApJL, 767, L8
Kushniruk, I., Bensby, T., Feltzing, S., et al. 2020, A&A, 638, A154
Leggett, S. K., Ruiz, M. T., & Bergeron, P. 1998, ApJ, 497, 294
Limoges, M. M., Bergeron, P., & Lépine, S. 2015, ApJS, 219, 19
Lu, Y. L., Angus, R., Curtis, J. L., David, T. J., & Kiman, R. 2021, AJ, 161, 189
Mamajek, E. E., & Hillenbrand, L. A. 2008, ApJ, 687, 1264
Marigo, P., Cummings, J. D., Curtis, J. L., et al. 2020, NatAs, 4, 1102
McCleery, J., Tremblay, P.-E., Gentile Fusillo, N. P., et al. 2020, MNRAS, 499, 1890
McKinney, W. 2010, in Proc. of the 9th Python in Science Conf., ed. S. van der Walt & J. Millman, 56
Mullan, D. J., & Houdebine, E. R. 2020, ApJ, 891, 128
Núñez, A., Agüeros, M. A., Covey, K. R., et al. 2022, ApJ, 931, 45
Pass, E. K., Charbonneau, D., Irwin, J. M., & Winters, J. G. 2022, ApJ, 936, 109
Paunzen, E., & Vanmunster, T. 2016, AN, 337, 239
Popinchalk, M., Faherty, J. K., Kiman, R., et al. 2021, ApJ, 916, 77
Reylé, C., Jardine, K., Fouqué, P., et al. 2021, A&A, 650, A201
Robertson, P., Stefansson, G., Mahadevan, S., et al. 2020, ApJ, 897, 125
Rojas-Ayala, B. 2023, arXiv:2301.03442
Rolland, B., Bergeron, P., & Fontaine, G. 2018, ApJ, 857, 56
See, V., Matt, S. P., Finley, A. J., et al. 2019, ApJ, 886, 120
Skumanich, A. 1972, ApJ, 171, 565
Soderblom, D. R. 2010, ARA&A, 48, 581
Spada, F., & Lanzafame, A. C. 2020, A&A, 636, A76
Strolger, L. G., Gott, A. M., Carini, M., et al. 2014, AJ, 147, 49
Subasavage, J. P., Jao, W.-C., Henry, T. J., et al. 2017, AJ, 154, 32
Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, NatMe, 17, 261
Yu, J., & Liu, C. 2018, MNRAS, 475, 1093
Zhang, S., Luo, A. L., Comte, G., et al. 2019, ApJS, 240, 31